Telco Customer Churn Classification

Data Audit, Splitting, and Training-Set Exploratory Analysis

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Abstract

This document reports the initial data workflow for the Telco Customer Churn classification project. It audits the raw file, validates the target definition, identifies missing or disguised missing values, checks identifier columns, applies deterministic data-quality corrections, constructs a held-out train-test split, and performs exploratory data analysis on the training set only. Feature-target patterns are inspected only after the split, so that modelling decisions are based on the development data while the held-out test set remains reserved for final evaluation.

Contents

1 Raw data audit and deterministic correction

1.1 Purpose of the raw audit

This section documents the first workflow stage of the Telco Customer Churn classification project. The purpose is to inspect the raw dataset before constructing the modelling table and before creating the held-out train-test split.

The raw audit is deliberately limited. It does not estimate predictive patterns, compare feature distributions by churn status, create churn-rate plots, select features, tune preprocessing, or fit models. Instead, it answers the following questions:

- Does the raw file load correctly?
- What are the raw columns and data types?
- Are there duplicate rows?
- Are there standard missing values?
- Are there disguised missing values, such as blank strings?
- Does the target column exist and contain valid labels?
- Is there an identifier column that should be excluded from modelling?
- Are there deterministic raw-data representation problems that must be corrected before splitting?

This separation is important for leakage control. The full raw file may be inspected for schema and data-quality problems, because a valid modelling dataset cannot be created without those checks. However, feature-target exploratory analysis and any data-dependent modelling choices are postponed until after the train-test split and will use the training set only.

1.2 Raw dataset overview

The raw dataset contains 7043 observations and 21 columns. No exact duplicate rows were found. Pandas did not detect any standard missing values.

Table 1: Raw dataset overview

Item	Value
Number of rows	7043
Number of columns	21
Duplicate rows	0
Total standard missing values	0
Columns with standard missing values	0

The absence of standard missing values is not enough to conclude that no information is missing. String columns may contain blank strings or placeholder values that are not detected by ordinary missing-value checks.

1.3 Missing-data decision framework

Before choosing a cleaning strategy, it is useful to distinguish two different missing-data problems:

1. Missing feature values: at least one input variable X is missing.
2. Missing target labels: the output variable Y is missing.

These two cases should not be handled in the same way. For missing feature values, the central production question is whether missing inputs may occur when the model is used in the future. If missing feature values may occur in production, the final model pipeline must be able to handle them. In that setting, missing values should remain present in validation and test data so that evaluation simulates production. Any imputation rule or preprocessing rule must be fitted on the training data only and then applied unchanged to validation, test, and production data.

If production data is expected to be clean but the training data contains missing feature values, the goal is different: the experiment should simulate clean production. In that case, possible training-data strategies include removing incomplete training rows, imputing training values, removing a feature, or using model-based imputation. The choice should be made using training and validation logic, not the held-out test set.

For missing target labels, the rules are different. Missing labels in the training set may require training only on labelled observations, label imputation, or semi-supervised learning. Missing labels in the test set should not simply be imputed and treated as known. If test labels are missing, performance should be reported with uncertainty, for example through best-case and worst-case bounds.

In this dataset, the raw audit found no missing target labels and no standard missing feature values. It did find a disguised missing or representation issue in `TotalCharges`, which is discussed below.

1.4 Disguised missing values

The string-column audit checks for blank strings after trimming whitespace. This is necessary because blank strings such as `""` or `" "` are not automatically treated as missing values by pandas.

Table 2: Blank-string audit for selected string-like columns

Column	Blank string count	Blank string percentage	Unique values
TotalCharges	11	0.16	6531
customerID	0	0.00	7043
PaymentMethod	0	0.00	4
MultipleLines	0	0.00	3
InternetService	0	0.00	3
OnlineSecurity	0	0.00	3
OnlineBackup	0	0.00	3

Only `TotalCharges` contains blank strings. This is important because `TotalCharges` is conceptually numeric, but the blank strings cause the raw column to be read as string-like.

1.5 Target-label validation

The target variable is `Churn`. The observed target labels are `No` and `Yes`. There are no missing target labels and no unexpected target labels.

Table 3: Target distribution in the raw dataset

Target level	Count	Percentage
No	5174	73.46
Yes	1869	26.54

The positive class is defined as `Churn = Yes`. The target distribution is moderately imbalanced, with approximately 26.54% churn. This check justifies using a stratified train-test split in the next workflow stage. It is not used for model selection.

Table 4: Target-label validity check

Item	Value
Expected negative label present	True
Expected positive label present	True
Missing target values	0
Unexpected target labels	None

1.6 Identifier check

The column `customerID` has 7043 unique values for 7043 rows. It is therefore a unique identifier candidate. The identifier is useful for traceability, but it should not be used as a modelling feature because it does not describe a generalizable customer characteristic.

Table 5: Identifier check

Column	Unique values	Number of rows	Unique value ratio
customerID	7043	7043	1.00
TotalCharges	6531	7043	0.93
MonthlyCharges	1585	7043	0.23
tenure	73	7043	0.01
PaymentMethod	4	7043	0.00

1.7 Investigation of `TotalCharges`

The blank-string audit showed 11 blank strings in `TotalCharges`. These rows were checked against `tenure`. The result is exact: the 11 blank `TotalCharges` rows are exactly the 11 rows where `tenure = 0`.

Table 6: Relationship between blank `TotalCharges` and zero tenure

Condition	Count
Blank `TotalCharges`	11
`tenure == 0`	11
Blank `TotalCharges` and `tenure == 0`	11
Blank `TotalCharges` and `tenure != 0`	0
Non-blank `TotalCharges` and `tenure == 0`	0

Converting `TotalCharges` to numeric creates exactly 11 missing values, which are the same 11 blank-string records.

Table 7: `TotalCharges` numeric conversion check

Item	Count
Standard missing before conversion	0
Missing after numeric conversion	11
New missing created by numeric conversion	11
Blank string count	11

The affected rows are all zero-tenure customers. Most of these records have a two-year contract and none of them churned in the raw data. The churn outcome is not used to justify the correction. The correction is justified by the meaning of `TotalCharges`: a zero-tenure customer has not accumulated charges yet.

1.8 Deterministic interim correction

The following correction is applied to an interim copy of the raw dataset:

1. Convert `TotalCharges` from string-like values to numeric values.
2. For rows where `tenure = 0` and `TotalCharges` is blank after conversion, set `TotalCharges = 0.0`.

This is not mean imputation, median imputation, model-based imputation, or a target-informed correction. It is a deterministic representation correction based on the definition of accumulated charges. The original raw dataframe is kept unchanged, and the corrected copy is saved as the interim dataset for the next workflow stage.

After the correction, `TotalCharges` is numeric, contains no missing values, and ranges from 0.0 to 8684.8.

Table 8: Post-correction check

Item	Value
Remaining total missing values	0
Remaining missing `TotalCharges`	0
`TotalCharges` dtype	float64
Minimum `TotalCharges`	0.00
Maximum `TotalCharges`	8684.80

1.9 Basic numeric domain-validity checks

Unusual values are not automatically errors. A high but possible value may be a natural observation that belongs to the production distribution. However, values that violate basic domain constraints are different: they may indicate coding errors, measurement errors, or corrupted records.

The three numeric variables in this dataset should be non-negative:

- `tenure` is the number of months a customer has stayed with the company;
- `MonthlyCharges` is the customer's monthly charge amount;
- `TotalCharges` is the accumulated charge amount.

A value of zero can be valid. For example, zero tenure occurs for newly registered customers, and the corresponding accumulated total charge can be zero. Therefore, this audit checks only for negative values. This is not statistical outlier detection. Statistical outlier analysis is postponed until training-set exploratory analysis.

Table 9: Numeric domain-validity check after interim correction

Feature	Minimum	Maximum	Negative count	Negative percentage
tenure	0.00	72.00	0	0.00
MonthlyCharges	18.25	118.75	0	0.00
TotalCharges	0.00	8684.80	0	0.00

No negative values were found in the numeric variables. Therefore, no additional corrupted numeric values are corrected in this raw-audit stage.

1.10 Output of this workflow stage

The raw audit produces an interim dataset with 7043 rows, 21 columns, and no missing values. The raw target column `Churn` remains unchanged. The next workflow stage will create the binary target `Churn_binary`, exclude `customerID` from the modelling feature set, and create the held-out stratified train-test split.

Table 10: Interim dataset save check

Item	Value
Number of rows	7043
Number of columns	21
Total missing values	0
Interim file	`data/interim/telco_churn_interim.csv`

2 Clean modelling dataset and train-test split

2.1 Purpose of this stage

The raw-data audit produced an interim dataset in which the representation problem in `TotalCharges` was corrected. The next step is to create the supervised modelling table and the held-out train-test split.

This stage is deliberately narrow. It creates the binary target, defines the feature set, validates semantic feature groups, and creates a stratified split. It does not perform exploratory feature-target analysis, one-hot encoding, scaling, feature engineering, resampling, feature selection, or model fitting. Those steps are postponed until after the split and should use the training set only.

2.2 Interim dataset

The interim dataset is loaded from:

`data/interim/telco_churn_interim.csv`.

It has the same raw columns as the original dataset, except that `TotalCharges` has been converted to a numeric representation and the zero-tenure blank values have been set to `0.0`. No remaining missing values are present at this stage.

Table 11: Interim dataset overview

Item	Value
Number of rows	7043
Number of columns	21
Duplicate rows	0
Total missing values	0
Columns with missing values	0

2.3 Binary target encoding

The supervised learning target is `Churn_binary`. The positive class is churn, because this is the event of practical interest.

$$\texttt{Churn\_binary} = \begin{cases} 1, & \text{if } \texttt{Churn = Yes}, \\ 0, & \text{if } \texttt{Churn = No}. \end{cases}$$

The encoding produces no missing values and preserves the two-class structure of the original target.

Table 12: Target encoding

Original target level	Encoded value
No	0
Yes	1

2.4 Modelling feature set

The modelling feature set excludes `customerID`, the original string target `Churn`, and the binary target `Churn_binary`. The identifier is excluded because it is row-specific and does not represent a generalizable customer characteristic. The original target is excluded because the model uses the encoded binary target.

The resulting modelling table contains 7043 observations, 19 feature columns, and one target column.

Table 13: Clean modelling table overview

Item	Value
Rows	7043
Feature columns	19
Target columns	1
Identifier included as feature	No
Original string target included as feature	No

2.5 Semantic feature groups

The 19 modelling features are divided into semantic groups. These groups describe the clean base dataset; they do not yet define the final preprocessing for every model. Model-specific preprocessing is handled later inside training-only pipelines.

Table 14: Feature groups for modelling

Feature group	Count	Features
Numeric features	3	`tenure`, `MonthlyCharges`, `TotalCharges`
Binary categorical features	6	`SeniorCitizen`, `gender`, `Partner`, `Dependents`, `PhoneService`, `PaperlessBilling`
Nominal categorical features	10	`MultipleLines`, `InternetService`, `OnlineSecurity`, `OnlineBackup`, `DeviceProtection`, `TechSupport`, `StreamingTV`, `StreamingMovies`, `Contract`, `PaymentMethod`

The feature-group validation confirms that every modelling feature belongs to exactly one group: no feature is missing from the groups, no grouped feature lies outside the modelling feature set, and no feature is assigned more than once.

2.6 Final checks before splitting

The final modelling table contains no missing values. Exact duplicate rows in the full interim data are absent. After excluding the identifier, duplicate feature-target combinations may exist, but this is not automatically a data-quality issue because multiple customers can genuinely share the same observed characteristics and churn label.

Table 15: Final checks before splitting

Item	Value
Missing values in modelling table	0
Duplicate full interim rows	0
Duplicate feature-target rows after excluding identifier	22

2.7 Target distribution before splitting

The target distribution is moderately imbalanced. The positive class, churn, represents approximately 26.5% of observations. This motivates a stratified train-test split. This target-only summary is used for split design rather than feature selection or model interpretation. No feature-target relationships are inspected before the train-test split.

Table 16: Target distribution before splitting

`Churn_binary`	Count	Percentage
0	5174	73.46
1	1869	26.54

2.8 Train-test split strategy

The clean modelling table is split into a training set and a held-out test set. The test set is reserved for final evaluation and should not be used for feature-target EDA, preprocessing design, model selection, hyperparameter tuning, or threshold tuning.

The split uses:

$$\texttt{test\_size} = 0.20, \qquad \texttt{random\_state} = 42.$$

The split is stratified by `Churn_binary`, so the train and test sets preserve the original churn proportion.

Table 17: Train-test split overview

Split	Rows	Percentage
Train	5634	79.99
Test	1409	20.01

2.9 Target distribution by split

The stratified split preserves the target distribution closely.

Table 18: Target distribution by split

Split	`Churn_binary`	Count	Percentage
Train	0	4139	73.46
Train	1	1495	26.54
Test	0	1035	73.46
Test	1	374	26.54

2.10 Saved outputs

The training and test datasets are saved locally as:

`data/processed/train.csv`, `data/processed/test.csv`.

These files form the boundary for the rest of the project. The next workflow stage should load only `train.csv` for exploratory analysis and modelling decisions. The held-out test file should remain unused until final model evaluation.

3 Exploratory data analysis

3.1 Purpose of this stage

The raw-data audit and splitting stages created a corrected interim dataset, a clean modelling table, and a held-out train-test split. This section performs exploratory data analysis using only the training set.

This restriction is methodological. Exploratory findings can influence later modelling choices, including feature engineering, transformations, rule-based baselines, model selection, outlier treatment, and interpretation. Therefore, feature distributions and feature-target relationships are not inspected on the held-out test set.

The goals of this stage are to understand the training data, identify modelling-relevant patterns, and produce report-ready figures and tables while preserving the test set for final evaluation. Large figures are split when needed for readability; a figure is kept in the main text when it is directly used for interpretation, while purely supplementary diagnostics can be moved to an appendix in later stages.

3.2 Training-set overview

The dataset used in this section contains 5634 observations. It has 19 feature columns and the binary target column `Churn_binary`. No missing values remain after the raw audit and deterministic `TotalCharges` correction.

Table 19: Training-set overview

Item	Value
Training rows	5634
Training columns	20
Missing values	0
Columns with missing values	0
Duplicate feature-target rows	15

The duplicate feature-target rows are not treated as a data-quality issue. The unique customer identifier was intentionally removed before modelling, so different customers can legitimately share the same observed feature values and churn label.

3.3 Target distribution

The positive class is churn, encoded as `1`. The target distribution remains moderately imbalanced, with churn representing 26.54% of observations.

Table 20: Target distribution

`Churn_binary`	Count	Percentage
0	4139	73.46
1	1495	26.54

This imbalance is not extreme, but it is large enough that accuracy alone will be an incomplete evaluation metric. Later modelling sections should include metrics that distinguish false positives from false negatives, such as precision, recall, specificity, F1-score, ROC-AUC, PR-AUC, and threshold analysis.

3.4 Numeric feature summary

The dataset has three numeric features: `tenure`, `MonthlyCharges`, and `TotalCharges`. Numeric summaries are computed on this development split only.

Table 21: Numeric feature summary

Feature	Count	Mean	Std.	Min	25%	Median	75%	Max
`tenure`	5634	32.49	24.57	0.00	9.00	29.00	55.00	72.00
`MonthlyCharges`	5634	64.93	30.14	18.40	35.66	70.50	90.00	118.75
`TotalCharges`	5634	2299.33	2279.20	0.00	402.98	1394.93	3835.83	8684.80

The numeric variables have different scales. This matters for later model families such as k-nearest neighbours, support vector machines, logistic regression with regularization, and neural networks, where scaling is usually required. Scaling should be fitted inside training-only pipelines rather than on the full dataset.

3.5 Numeric feature distributions

Figure 1 shows histograms and boxplots for the three numeric features in this split.

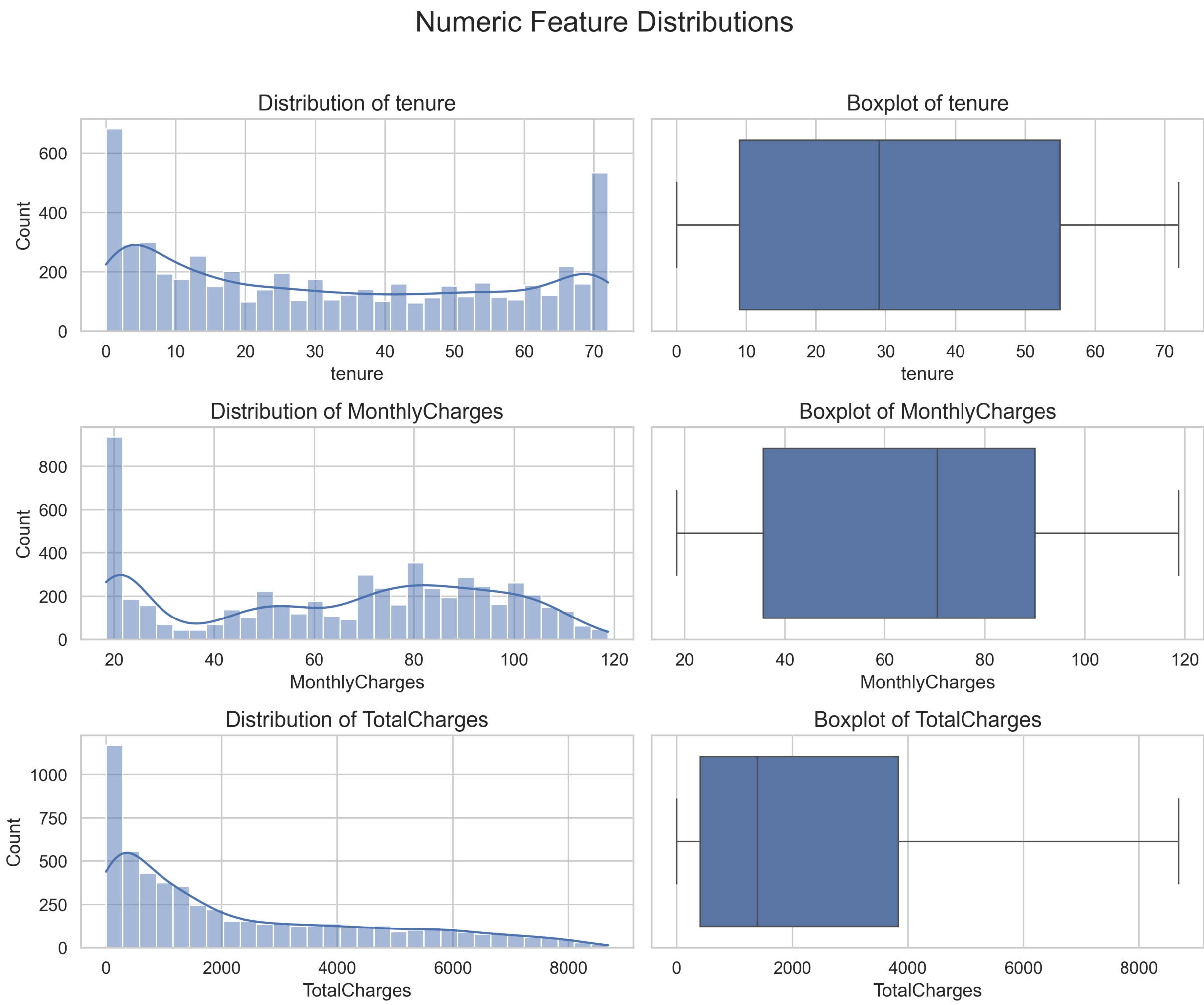

Figure 1: Numeric feature distributions and boxplots

The distribution of `tenure` has substantial mass near low values and also many observations near the upper end of the observed range. This indicates that the training set contains both newly joined customers and long-tenure customers.

The distribution of `MonthlyCharges` is not symmetric. There is a large group of customers with relatively low monthly charges and a broad group at higher monthly charge levels. The distribution of `TotalCharges` is strongly right-skewed. This is expected because total charges accumulate over time: many customers have low accumulated charges, while long-tenure or high-monthly-charge customers can have much larger accumulated totals.

These plots are descriptive. They do not justify automatic outlier removal. The raw audit already checked for impossible negative numeric values. High but plausible values should be treated as potential natural observations unless there is separate evidence that they are coding errors.

3.6 Numeric feature distributions by churn

Figure 2 compares the numeric feature distributions between churners and non-churners.

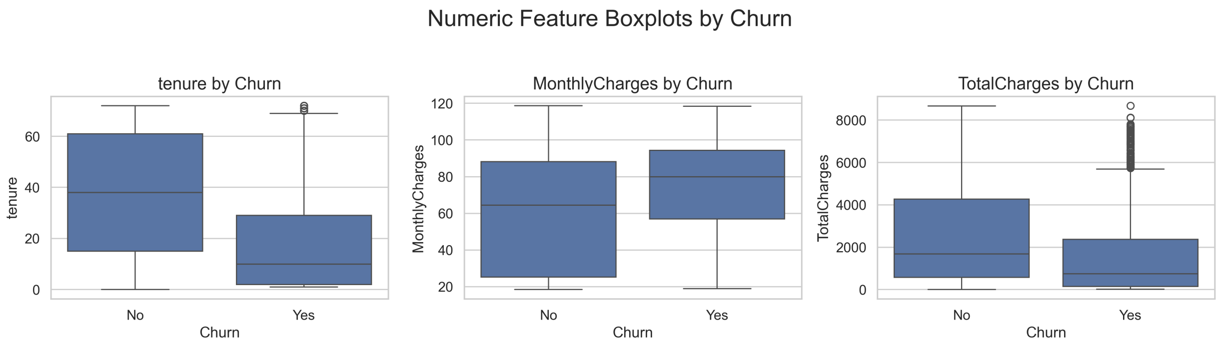

Figure 2: Numeric feature boxplots by churn

The class-conditioned boxplots show clearer feature-target structure than the marginal distributions. Customers who churned have substantially lower `tenure` than customers who did not churn. This is one of the strongest numeric patterns in the training data and suggests that newer customers are more likely to leave.

`MonthlyCharges` tends to be higher for churners. This indicates that customers with higher monthly bills may be at greater churn risk, at least as a marginal training-set association. `TotalCharges` tends to be lower for churners, but this should be interpreted carefully. Since `TotalCharges` accumulates over time, lower total charges among churners are consistent with churners having shorter tenure.

3.7 Numeric scatter matrix

Figure 3 shows a scatter matrix for the numeric features, coloured by churn status.

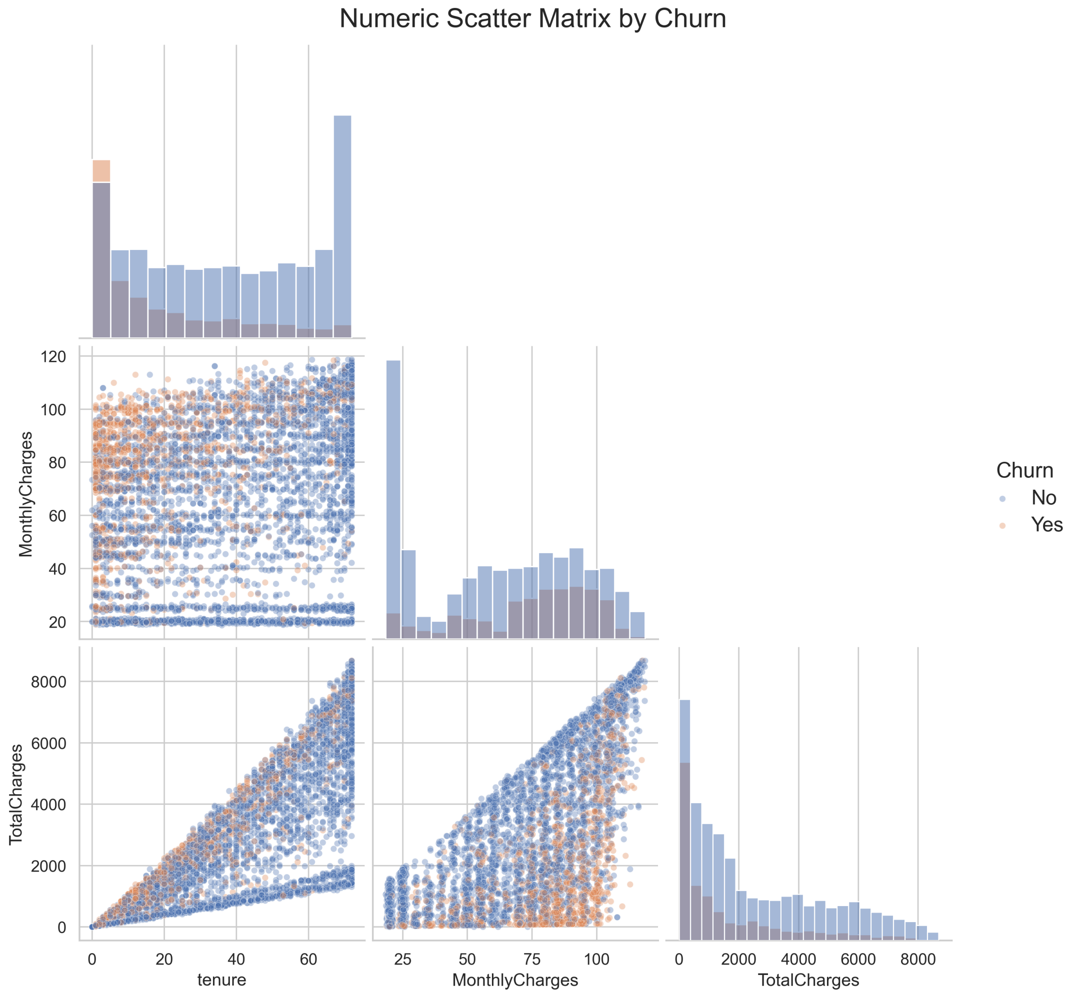

Figure 3: Numeric scatter matrix by churn

The scatter matrix shows visible churn-related patterns, but it does not show a clean visual separation between churners and non-churners using any single pair of numeric variables. Churners are more concentrated among customers with shorter tenure and are also relatively visible among customers with higher monthly charges. However, the two classes overlap substantially across the numeric feature space, suggesting that churn is not separable by a simple pairwise numeric boundary.

The clearest structural pattern is the relationship between `tenure` and `TotalCharges`. The triangular shape is expected because total charges accumulate over months and are also affected by monthly charges. For a fixed tenure, customers with higher monthly charges can accumulate higher total charges. This motivates using models that can combine numeric and categorical information, and possibly models that can learn interactions.

3.8 Numeric correlation matrix

Figure 4 shows the Pearson correlation matrix among numeric features and `Churn_binary` in this split.

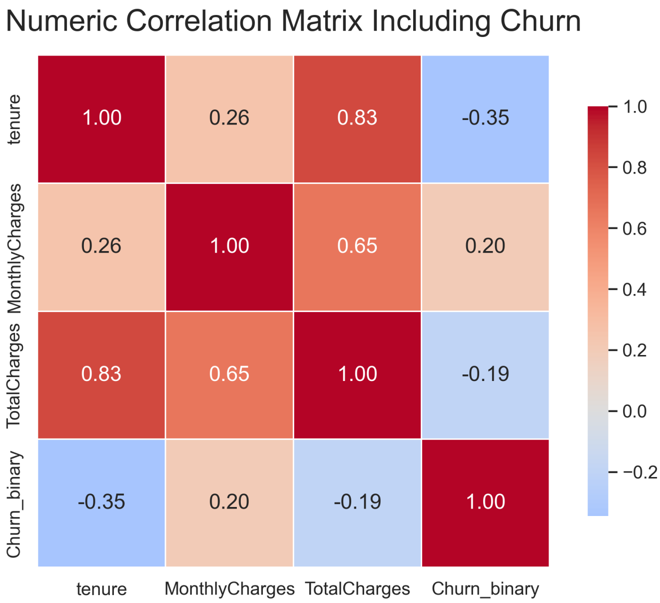

Figure 4: Numeric correlation matrix including `Churn_binary`

The strongest numeric correlation with churn is for `tenure`, with correlation approximately -0.35. Longer-tenure customers are less likely to churn in the training data. `MonthlyCharges` has a positive correlation with churn of approximately 0.20, while `TotalCharges` has a negative correlation with churn of approximately -0.19.

The correlation matrix also shows that `tenure` and `TotalCharges` are strongly correlated, approximately 0.83, and that `MonthlyCharges` and `TotalCharges` are also positively correlated, approximately 0.65. These are marginal linear associations only. They do not capture nonlinear relationships, interactions, or the effects of categorical variables.

3.9 Categorical feature frequencies

Figures 5 and 6 show the distribution of categorical feature levels. The categorical variables are split across two figures to keep axis labels and category names readable in the compiled report.

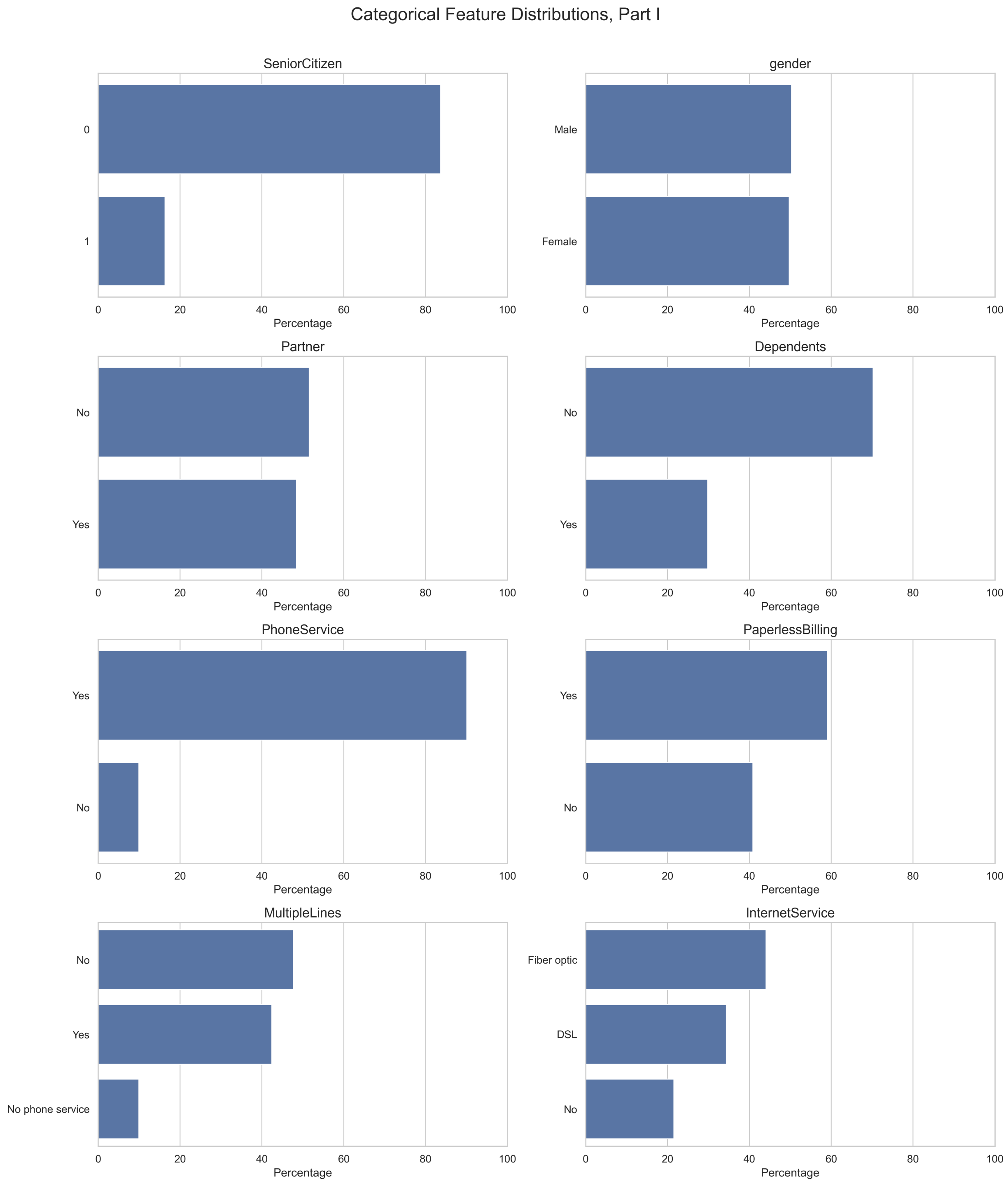

Figure 5: Categorical feature distributions, Part I

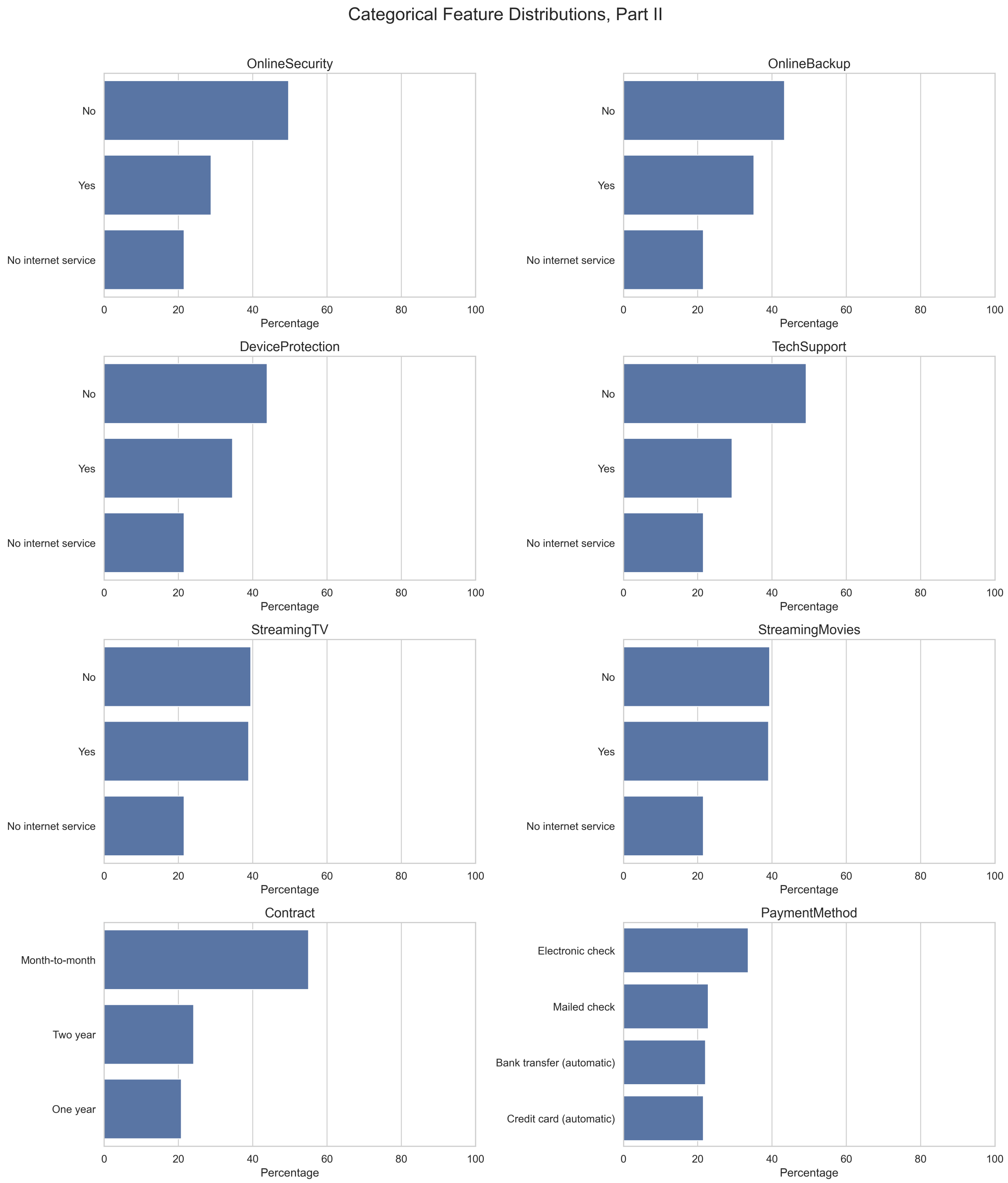

Figure 6: Categorical feature distributions, Part II

Several categorical predictors contain strongly imbalanced levels. For example, `PhoneService = Yes` represents 90.08% of the training set, `SeniorCitizen = 0` represents 83.67%, and `Contract = Month-to-month` represents 55.06%. These imbalances matter because rare levels can create unstable estimates in simple baselines and can affect how much information a model can learn from a category.

Several internet-service add-on variables contain the structural level `No internet service`. This is not missingness. It is an informative category indicating that the customer does not have internet service. Later preprocessing should preserve this category rather than treating it as a missing value.

3.10 Categorical churn rates

Figures 7 and 8 show the churn rate within each categorical level. Both parts are included in the main section because they support the interpretation: Part I contains demographic, household, billing, phone-service, and internet-service variables, while Part II contains several service add-ons as well as contract type and payment method.

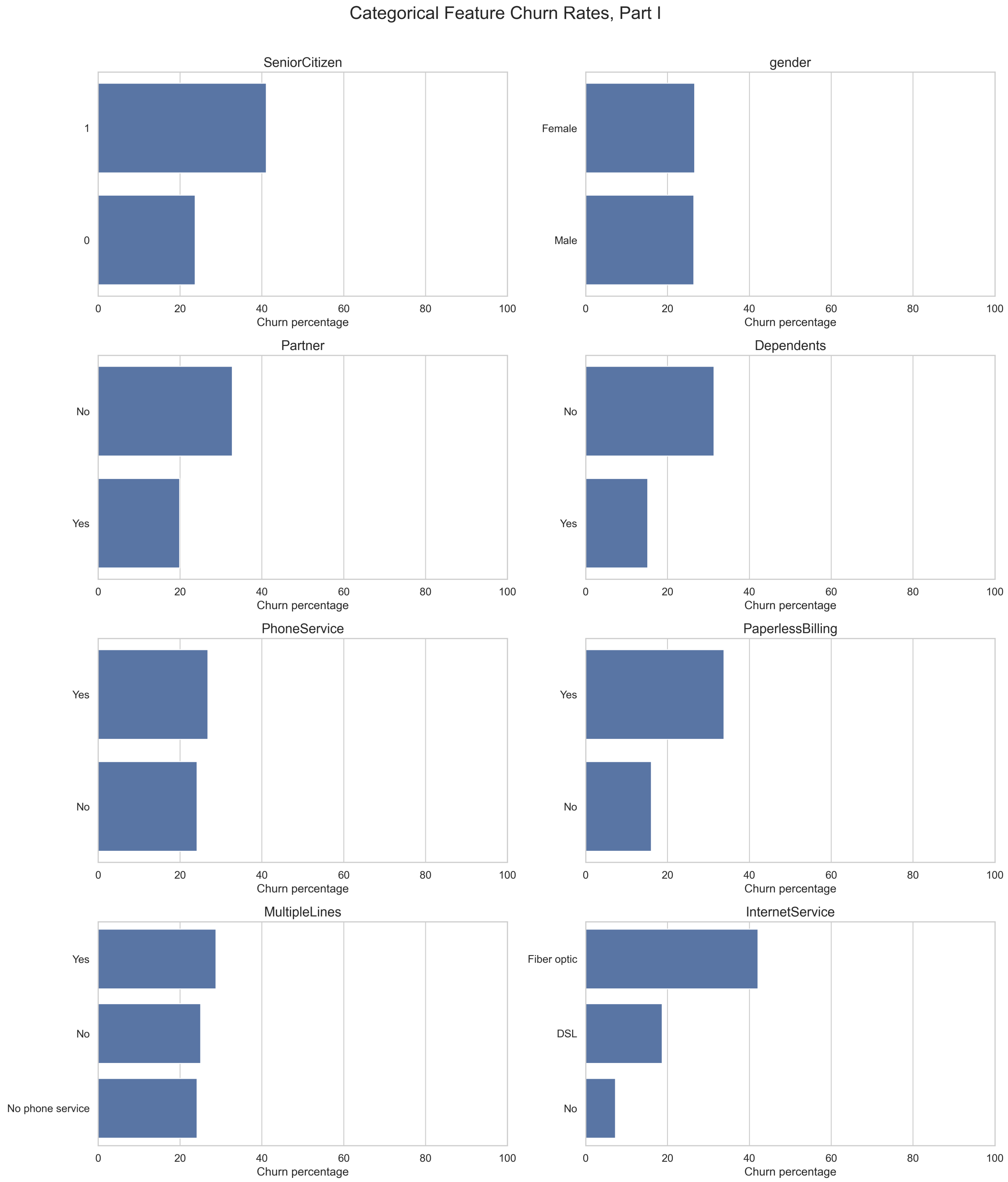

Figure 7: Categorical feature churn rates, Part I

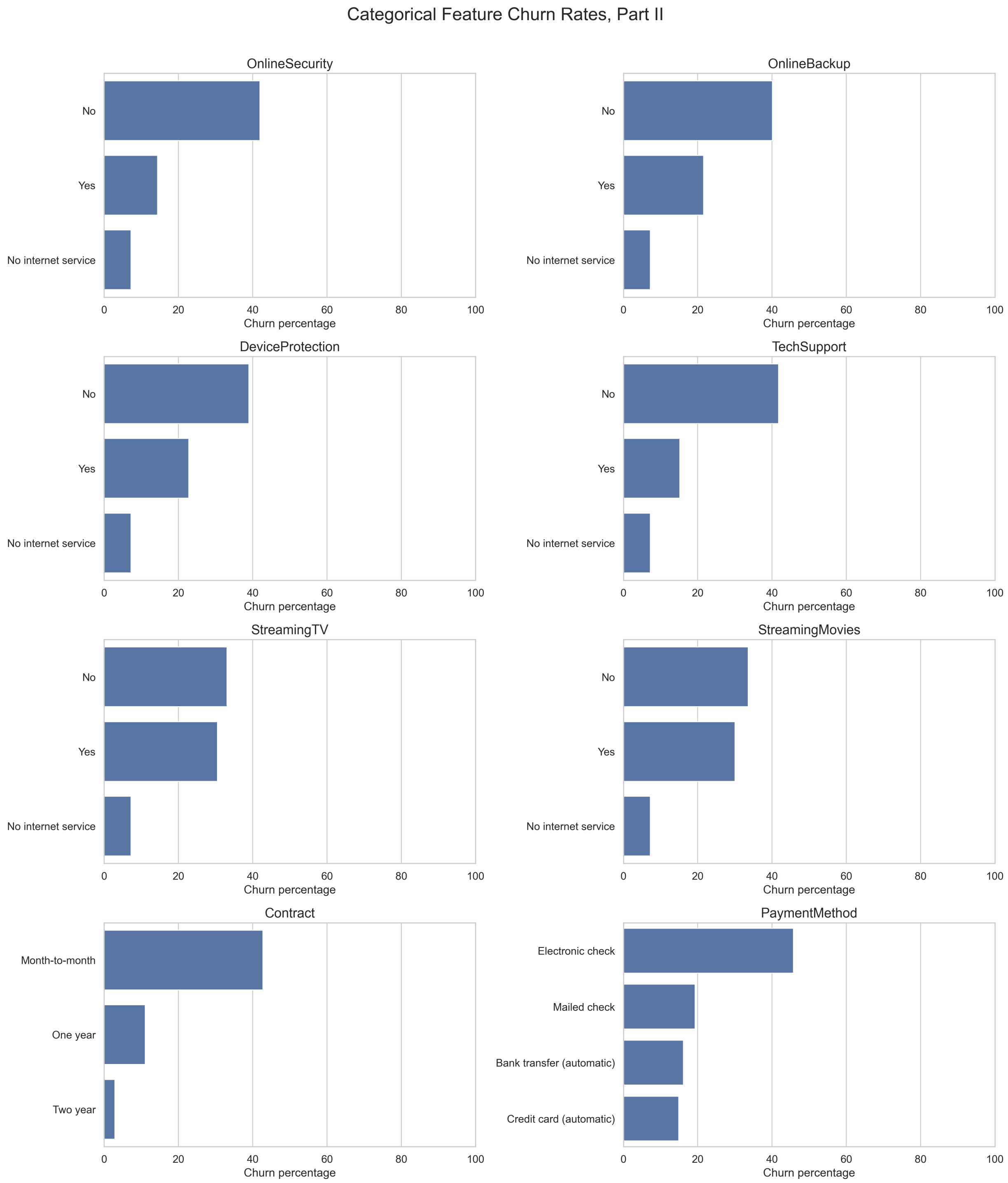

Figure 8: Categorical feature churn rates, Part II

The categorical churn-rate plots reveal several large training-set differences. Contract type is one of the strongest categorical patterns: month-to-month customers have a churn rate of 42.75%, compared with 11.08% for one-year contracts and 2.87% for two-year contracts. Internet service type also differs strongly: fiber optic customers have a churn rate of 42.09%, compared with 18.69% for DSL and 7.25% for customers with no internet service.

Payment method is also informative. Customers using electronic check have a churn rate of 45.74%, higher than mailed check at 19.28%, bank transfer at 16.16%, and credit card at 14.92%. Customers without online security or tech support also show much higher churn rates than customers with those services.

Some variables show weak marginal separation. For example, churn rates for `gender` are almost identical, and `PhoneService` shows only a small difference. This does not mean these variables must be removed, because they may still contribute through interactions or in combination with other features. It does suggest that their standalone marginal relationship with churn is weak.

3.11 Selected categorical churn-rate table

Table 22 reports selected categorical levels that are especially relevant for interpretation and later baseline design.

Table 22: Selected categorical churn rates

Feature	Level	Count	Training %	Churn %
`Contract`	Month-to-month	3102	55.06	42.75
`Contract`	One year	1173	20.82	11.08
`Contract`	Two year	1359	24.12	2.87
`InternetService`	Fiber optic	2483	44.07	42.09
`InternetService`	DSL	1937	34.38	18.69
`InternetService`	No	1214	21.55	7.25
`PaymentMethod`	Electronic check	1891	33.56	45.74
`PaperlessBilling`	Yes	3331	59.12	33.80
`PaperlessBilling`	No	2303	40.88	16.02
`SeniorCitizen`	1	920	16.33	41.09
`SeniorCitizen`	0	4714	83.67	23.70

These are associations in the training data, not causal effects. They are useful for understanding the dataset, designing simple baselines, and interpreting later models.

3.12 Summary

The EDA suggests that the dataset contains meaningful predictive structure. Numerically, churn is most clearly associated with lower tenure and somewhat higher monthly charges. `TotalCharges` is lower among churners, but this is closely related to the shorter tenure of churned customers.

Categorically, the strongest visible churn-rate differences occur for contract type, internet service type, payment method, online security, tech support, paperless billing, and senior-citizen status. These patterns motivate the next stage: building preprocessing pipelines and baseline classifiers using only the training set.

References

[1] Vrije Universiteit Amsterdam Machine Learning course notes. *Data pre-processing, Part 1: Missing values and outliers.* Used for the principles of not taking data at face value, distinguishing missing feature values from missing target labels, considering the production setting, and distinguishing corrupted values from natural extreme observations.

[2] Vrije Universiteit Amsterdam Machine Learning course notes. *Model Evaluation, Part 1: Experiments.* Used for the principle that the held-out test set should not be reused during model selection, preprocessing design, feature engineering, or hyperparameter tuning.